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Manufacture and use of allogrooming tools by wild killer whales

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The manufacture and use of tools, while widespread in terrestrial animals[1], has been less frequently reported in marine taxa[2]. In cetaceans, clear examples of tool use are largely restricted to foraging contexts, with no reports of cetaceans fashioning tools by modifying objects. Here, we report evidence of the widespread manufacture and use of allogrooming tools in a population of resident killer whales (*Orcinus orca ater*).

From 10th April to 27th July, 2024, we conducted observations of southern resident killer whales from an unoccupied aerial vehicle in the central Salish Sea (Figure S1A). We observed whales fashioning short lengths of bull kelp (*Nereocystis luetkeana*) stipe from complete stalks, positioning the stipe between themselves and a partner, and then rolling the kelp along their bodies (Figure 1A–C). Cetaceans are often observed with kelp draped over their bodies, a behavior called 'kelping'[3]. This newly described behavior, in which whales kelp with the aid of a social partner (and which may be a social elaboration on kelping behavior), we will thus call 'allokelping'.

We documented 30 bouts of allokelping across 8 out of 12 days of observation (Supplemental information). Allokelping occurred across all age-sex classes and all three major social groups (J, K and L pods). Within J pod (nine total hours of footage over 11 days), allokelping occurred across all six matrilines and 19 out of 25 individuals. Observations were sparser in K and L pods due to much lower sampling intensity (K pod: one hour across two days; L pod: 21 minutes on one day).

We observed whales acquiring stipe segments for allokelping on eight occasions. On seven of these occasions, the length of stipe was fashioned from an intact stalk; in the remaining observation the whale retrieved a suitable segment from a floating kelp mat. Whales fashion lengths of kelp stipe for allokelping by using their teeth to grab a kelp stalk by the narrow end of the stipe near the holdfast. They then use the kelp's drag, the motion of their body and their teeth to separate a short length of the kelp stipe (Figure 1A and Supplemental information).

After fashioning a length of kelp, the whale will then approach a partner, flip the length of kelp onto their rostrum, and press their head and the kelp against their partner's flank. The two whales then maneuver to keep the kelp between them while rolling it across their bodies (Figure 1B and Supplemental information). Whales maintained continuous contact for a median of 32 seconds during allokelping (IQR = 38; maximum = 186). During contact, whales roll and twist their bodies, often adopting an exaggerated S-shaped posture (Figure 1C).

When contact is broken, the whale in the rear may retrieve the kelp and re-initiate contact (Supplemental information). We observed 28 re-initiations in 13 separate bouts. Pairs of whales often switched which individual initiated contact within a bout; role switching occurred in 66% of bouts in which multiple initiations were observed. Observations of allokelping bouts lasted for a median of one minute, 19 seconds (IQR = 1.35 minutes), with the longest observation lasting 12 minutes, 4 seconds.

Allokelping is dissimilar to object-oriented play previously reported in delphinids. Allokelping whales did not exhibit play chases, surface active behaviors or play 'keep-away' with the piece of kelp[4]. The highly stereotyped nature of this behavior, combined with its occurrence across all age-sex classes, further suggests that allokelping is not purely a form of object-oriented play[4].

We hypothesize that allokelping is similar to skin maintenance behaviors exhibited by other cetaceans, where exaggerated S-poses are used to achieve contact with exfoliating surfaces (e.g. stones or sand)[5]. Unlike southern resident killer whales,

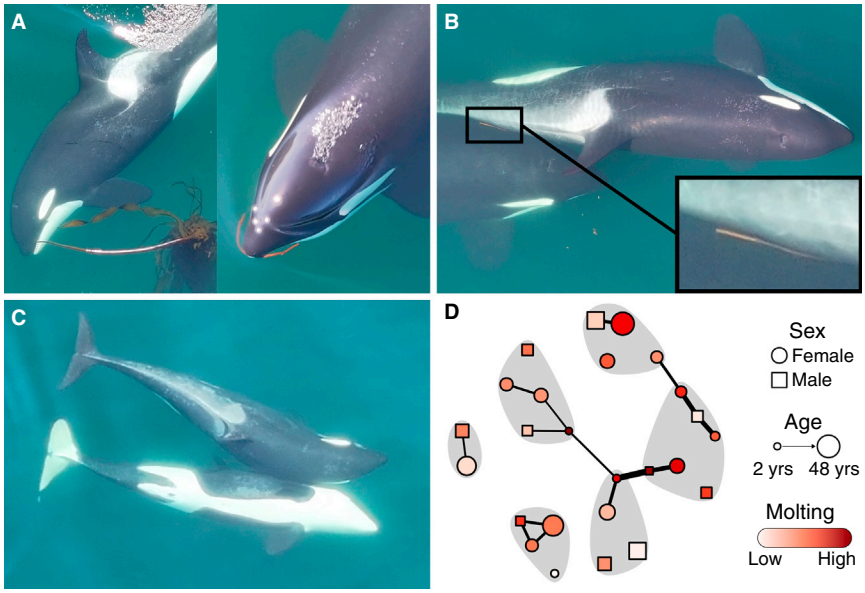


Figure 1. Allokelping in southern resident killer whales.
(A) Fashioning kelp stipes for allokelping. (Left) J41 grasps a kelp stalk by the holdfast end and (right) detaches a short segment. (B) J19 and J51 allokelping with the kelp stipe visible between them (inset). (C) J56 exhibiting an *S*-pose while allokelping with J57 (inverted). (D) J pod allokelping network. Nodes (circles/squares) represent individuals, while edge (lines) thickness indicates the observed rate of allokelping. Grey polygons demarcate matrilines. Node size indicates age, shape indicates sex, and color indicates relative level of skin molting as indicated in the legend. Aerial images collected under NMFS research permit 27038.

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northern resident and some Alaskan resident killer whales rub themselves on smooth pebble beaches[6], though the function of this behavior is not well established. Allokelping may be a form of social skin maintenance, making it an example of tool-assisted allogrooming. Social skin maintenance (e.g. flipper-to-body contact to remove dead skin[7]) has been reported in cetaceans; however, tool use for this purpose has not been documented in this taxon. Reports of tool-assisted allogrooming are largely limited to primates, where observations are rare and typically occur in captivity[8].

Allogrooming is often a key affiliative behavior that serves both hygienic and social functions[9]. Despite a relatively small sample size (27 bouts across 25 individuals), we conducted an initial analysis of allokelping rates within J pod to explore this hypothesis (Figure 1D and Supplemental information including Table S1). Like other affiliative social behaviors in this population[10], allokelping is positively correlated with maternal kinship ($\beta \pm SE = 0.91 \pm 0.21$, posterior $P(\beta > 0) > 0.99$) and age similarity ($\beta \pm SE = 0.90 \pm 0.45$, posterior $P(\beta > 0) = 0.98$). There is also tentative evidence that allokelping is positively correlated with levels of skin molting ($\beta \pm SE = 0.43 \pm 0.33$, posterior $P(\beta > 0) = 0.93$; see supplemental materials including Figure S1B), as would be expected if allokelping aided in the removal of dead skin. Unlike play and other forms of affiliative physical contact[4,10], we found no evidence that allokelping rates decreased with age ($\beta \pm SE = -0.01 \pm 0.34$, posterior $P(\beta < 0) = 0.53$).

A robust test of the hypothesized functions of allokelping requires data-rich analyses of how this behavior impacts other social interactions and relates to skin condition, and extension of this analysis to the entire population. If a functional benefit of allokelping can be confirmed, this would expand the known taxonomic breadth and contexts of cetacean tool use, and would be the first case of tool manufacture (in this case using the 'detach' mode[1]) in marine mammals. Furthermore, allokelping may be the first example of tool use in a wild animal benefiting two individuals simultaneously, and the first case of non-human animals manipulating a tool with the core of their body rather than an appendage.

We hypothesize that allokelping is a cultural behavior unique to southern resident killer whales. Future work should investigate if and how allokelping is learned, and whether it occurs in other killer whale societies. The southern resident killer whales are critically endangered, and maintaining their 'cultural continuity' is a key goal under Canada's Species at Risk Act. Declining bull kelp in southern resident killer whale habitat due to increasing temperatures may pose a threat to the persistence of this behavior. The beach-rubbing behavior of northern resident killer whales is the target of conservation action due to its cultural significance; allokelping may warrant similar measures.

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AUTHOR CONTRIBUTIONS

All authors conceived of and designed the study, with initial identification of allokelping by M.N.W. and R.E.J. M.N.W., R.E.J., C.A.G., A.C.-R., A.A.V., T.A.R., and D.K.E. carried out fieldwork. M.N.W., R.E.J., C.A.G., A.C.-R., A.A.V., and T.A.R. curated the dataset. M.N.W. and R.E.J. performed video analysis. C.A.G. and D.K.E. contributed to scoring molting levels for individual killer whales. M.N.W. conducted statistical analysis. M.N.W. wrote the initial draft of the manuscript, with subsequent input from all authors. M.N.W. and D.P.C. led funding acquisition.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

Supplemental information including one figure, one table, methods, link to videos of allokelping and kelp fashioning can be found with this article online at https://doi.org/10.1016/j.cub.2025.04.021.

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